1. The number of real solutions of; $\cos^{-1} x + \cos^{-1} 2x = -\pi$ is :

(A) 0 (B) 1 (C) 2 (D) infinitely many

$\cos^{-1} x + \cos^{-1} 2x = -\pi$ के वास्तविक हलों की संख्या है :

(A) 0 (B) 1 (C) 2 (D) अनन्त हल

Ans. A

Sol. $\cos^{-1} x + \cos^{-1} 2x = -\pi$

$[0, \pi]$ $[0, \pi]$

$[0, 2\pi]$ = $-\pi$

+ve $-$ve

Not possible

2. If $\sum_{i=1}^{2n} \cos^{-1} x_i = 0$, then find the value of $\sum_{i=1}^{2n} x_i$:

यदि $\sum_{i=1}^{2n} \cos^{-1} x_i = 0$, तब $\sum_{i=1}^{2n} x_i$ का मान होगा :

(A) 0 (B) n (C) 2n (D) $\frac{n(n+1)}{2}$

Ans. C

Sol. $\cos^{-1}x_1 + \cos^{-1}x_2 + \cos^{-1}x_3 + ...+ \cos^{-1} x_m = 0$

$[0, \pi]$	$[0, \pi]$	$[0, \pi]$	$[0, \pi]$
1	1	1	1
0	0	0	0
$x_1 = 1$	$x_2 = 1$	$x_3 = 1$	$x_m = 1$

$\sum_{i=1}^{2n} = 1+1+1+.....+1 = 2n$

n terms

3. If α, β are the roots of the equation; $6x^2 + 11x + 3 = 0$ $[\beta > \alpha]$ then :

(A) both $\cos^{-1} \alpha$ and $\cos^{-1} \beta$ are real (B) both $\cos^{-1} \alpha$ and $\text{cosec}^{-1} \beta$ are real

(C) both $\cot^{-1} \alpha$ and $\cos^{-1} \beta$ are real (D) none of these

यदि α, β समीकरण $6x^2 + 11x + 3 = 0$ $[\beta > \alpha]$ के मूल है तब :

(A) $\cos^{-1}\alpha$ एवं $\cos^{-1}\beta$ दोनों वास्तविक है (B) $\cos^{-1}\alpha$ एवं $\text{cosec}^{-1}\beta$ दोनों वास्तविक है

(C) $\cot^{-1}\alpha$ एवं $\cos^{-1}\beta$ दोनों वास्तविक है (D) इनमें से कोई नहीं

Ans. C

Sol. $6x^2 + 11x + 3 = 0$

$6x^2 + 9n + 2x + 3 = 0$

$3x(2x + 3) + (2x + 3) = 0$

$$x = \frac{-1}{3}, -\frac{3}{2}$$

$$\beta = -\frac{1}{3},\ \alpha = -\frac{3}{2}$$

4. The set of values of 'a' for which equation $x^2 + ax + \sin^{-1}(x^2 - 4x + 5) + \cos^{-1}(x^2 - 4x + 5) = 0$ has at least one solution, is :

(A) $\left(-\infty, -\sqrt{2\pi}\right] \cup \left[\sqrt{2\pi}, \infty\right)$ (B) $\left(-\infty, -\sqrt{2\pi}\right) \cup \left(\sqrt{2\pi}, \infty\right)$

(C) R (D) none of these

'a' के मानों का समुच्चय जिसके लिए समीकरण $x^2 + ax + \sin^{-1}(x^2 - 4x + 5) + \cos^{-1}(x^2 - 4x + 5) = 0$ का कम से कम एक हल अस्तित्व में हो :

(A) $\left(-\infty, -\sqrt{2\pi}\right] \cup \left[\sqrt{2\pi}, \infty\right)$ (B) $\left(-\infty, -\sqrt{2\pi}\right) \cup \left(\sqrt{2\pi}, \infty\right)$

(C) R (D) इनमें से कोई नहीं

Ans. D

Sol. $x^2 + ax + \sin^{-1}(x^2 - 4x + 5) + \cos^{-1}(x^2 - 4x + 5) = 0$

$$-1 \le \underset{\cup}{x^2 - 4x + 5} \le 1$$

$$\frac{-\delta}{4x} = \frac{-(16 - 2x)}{4}$$

$\Rightarrow 1$

$x = 1$

$$1 + a + \frac{\pi}{2} = 0 \Rightarrow a = -\left(1 + \frac{\pi}{2}\right)$$

5. If $-1 \le x, y, z \le 1$, such that $\sin^{-1} x + \sin^{-1} y + \sin^{-1} z = \frac{3\pi}{2}$, then find the value of

$x^{2000} + y^{2001} + z^{2002} - \frac{9}{x^{2000} + y^{2001} + z^{2002}}$:

यदि $-1 \le x, y, z \le 1$ तथा $\sin^{-1} x + \sin^{-1} y + \sin^{-1} z = \frac{3\pi}{2}$, तब $x^{2000} + y^{2001} + z^{2002} - \frac{9}{x^{2000} + y^{2001} + z^{2002}}$ का मान होगा:

(A) 0 (B) 1 (C) 2 (D) 3

Ans. A

Sol. $\sin^{-1}x + \sin^{-1}y + \sin^{-1}z = \frac{3\pi}{2}$

$\downarrow \quad \downarrow \quad \downarrow$

$\frac{\pi}{2} \quad \frac{\pi}{2} \quad \frac{\pi}{2}$

$x = 1, \quad y = 1, \quad z = 1$

$\Rightarrow x = y = z = 1 \Rightarrow x^{2000} + y^{2001} + z^{2002} - \frac{9}{x^{2000} + y^{2001} + z^{2002}}$

$1 + 1 + 1 - \frac{9}{1+1+1} = 3 - 3 = 0$

6. The principal value of $\cos^{-1}\left(-\sin\frac{7\pi}{6}\right)$ is :

(A) $\frac{5\pi}{3}$ (B) $\frac{7\pi}{6}$ (C) $\frac{\pi}{3}$ (D) none of these

$\cos^{-1}\left(-\sin\frac{7\pi}{6}\right)$ का मुख्य मान होगा :

(A) $\frac{5\pi}{3}$ (B) $\frac{7\pi}{6}$ (C) $\frac{\pi}{3}$ (D) इनमें से कोई नहीं

Ans. C

Sol. $\cos^{-1}\left(-\sin\frac{7\pi}{6}\right)$

$= \cos^{-1}\left(-\sin\left(\pi+\frac{\pi}{6}\right)\right) = \cos^{-1}\left(\sin\frac{\pi}{6}\right)$

$= \cos^{-1}\left(\frac{1}{2}\right) = \frac{\pi}{3}$

7. The domain of $\sqrt{\sec^{-1}\left(\frac{2-|x|}{4}\right)}$ is :

$\sqrt{\sec^{-1}\left(\frac{2-|x|}{4}\right)}$ का प्रान्त होगा :

(A) R (B) R – (–1, 1) (C) R – (– 3, 3) (D) R – (–6, 6)

Ans. D

Sol. $\sqrt{\sec^{-1}\left(\frac{2-|x|}{4}\right)}$

$\frac{2-|x|}{4} \geq 1$ $\qquad$ $\frac{2-|x|}{4} \leq -1$

$2-|x| \geq 4$ $\qquad$ $2|x| \leq -4$

$|x| \leq -2$ $\qquad$ $|x| \geq 6$

$x \in R - (-6, 6)$

8. The domain of the function $\sec^{-1}[x^2 - x + 1]$, is given by : (where $[\cdot]$ is greatest integer function) :

(A) [0, 1] (B) $(-\infty, 0] \cup [1, \infty)$ (C) $\left[\frac{1-\sqrt{5}}{2}, \frac{1+\sqrt{5}}{2}\right]$ (D) None of these

फलन $\sec^{-1}[x^2 - x + 1]$, का प्रान्त होगा : (जहाँ $[\cdot]$ महत्तम पूर्णांक फलन दर्शाता है) :

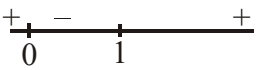
(A) [0, 1] (B) $(-\infty, 0] \cup [1, \infty)$ (C) $\left[\frac{1-\sqrt{5}}{2}, \frac{1+\sqrt{5}}{2}\right]$ (D) इनमें से कोई नहीं

Ans. B

Sol. $[x^2 - x + 1] \geq 1$ $\quad$ $[x^2 - x + 1] \leq 1$

$x^2 - x + 1 \geq 1$ $\quad$ $x^2 - x + 1 \leq 0$

$x(x-1) \geq 0$ $\quad$ $x \in \phi$



$x \in (-\infty, 0] \cup [1, \infty)$

9. The domain of definition of the function $f(x) = \frac{\cot^{-1} x}{\sqrt{\{x^2 - [x^2]\}}}$, where $[x]$ denotes the greatest integer less than or equal to x is :

फलन $f(x) = \frac{\cot^{-1} x}{\sqrt{\{x^2 - [x^2]\}}}$ का प्रान्त होगा, (जहाँ $[\cdot]$ महत्तम पूर्णांक फलन दर्शाता है) :

(A) R $\quad$ (B) $R - \{\pm \sqrt{n} : n \in I^+ \cup \{0\}\}$

(C) $R - \{0\}$ $\quad$ (D) $R - \{n : n \in I\}$

Ans. B

Sol. $(x) = \frac{\cot^{-1} x}{\sqrt{\{x^2 - [x^2]\}}}$

$= \frac{\cot^{-1}(x)}{\sqrt{\{x^2\}}}$

$R - \{\pm \sqrt{n} : n \in I^+ \cup \{0\}\}$

10. If D is the domain of the function $f(x) = \sqrt{1-2x} + 3\sin^{-1}x$ then D is equal to :

यदि D फलन $f(x) = \sqrt{1-2x} + 3\sin^{-1}x$ का प्रान्त है तो D होगा :

(A) $\left[-1, \frac{1}{2}\right]$ $\quad$ (B) $\left[-\frac{1}{3}, 0\right]$ $\quad$ (C) $\left[-\frac{1}{3}, 1\right]$ $\quad$ (D) $\left[\frac{1}{2}, 1\right]$

Ans. A

Sol. $f(x) = \sqrt{1-2x} + 3\sin^{-1}x$

$1 - 2x \geq 0 \quad x \in [-1, 1]$

$2x - 1 \leq 0$

$\left[-1, \frac{1}{2}\right]$

11. Let $f(x) = \sec^{-1} x + \tan^{-1} x$. Then $f(x)$ is real for :

(A) $x \in [-1, 1]$ (B) $x \in R$

(C) $x \in (-\infty, -1] \cup [1, \infty)$ (D) None of these

माना $f(x) = \sec^{-1} x + \tan^{-1} x$ तब $f(x)$ वास्तविक होगा यदि :

(A) $x \in [-1, 1]$ (B) $x \in R$

(C) $x \in (-\infty, -1] \cup [1, \infty)$ (D) इनमें से कोई नहीं

Ans. C

Sol. $f(x) = \sec^{-1} x + \tan^{-1} x$

$|x| \geq 1$ $x \in R$



$|x| \geq 1$

$x \in (-\infty, -1] \cup [1, \infty)$

12. Range of $f(x) = \sin^{-1}x + \tan^{-1}x + \sec^{-1}x$ is :

(A) $\left(\frac{\pi}{4}, \frac{3\pi}{4}\right)$ (B) $\left[\frac{\pi}{4}, \frac{3\pi}{4}\right]$ (C) $\left\{\frac{\pi}{4}, \frac{3\pi}{4}\right\}$ (D) None of these

फलन $f(x) = \sin^{-1}x + \tan^{-1}x + \sec^{-1}x$ का परिसर होगा :

(A) $\left(\frac{\pi}{4}, \frac{3\pi}{4}\right)$ (B) $\left[\frac{\pi}{4}, \frac{3\pi}{4}\right]$ (C) $\left\{\frac{\pi}{4}, \frac{3\pi}{4}\right\}$ (D) इनमें से कोई नहीं

Ans. C

Sol. $f(x) = \sin^{-1}x + \tan^{-1}x + \sec^{-1}x$

$\downarrow \quad \downarrow \quad \downarrow$

$[-1, 1] \quad R \quad (-\infty - 1] \cup [1, \infty]$

$x = -1 \qquad x = 1$

$f(x) = \sin^{-1}(-1) + \tan^{-1}(-1) + \sec^{-1}(-1)$

$\Rightarrow \frac{-\pi}{2} + \frac{-\pi}{4} + \pi + 0$

$= \pi - \frac{-3\pi}{4} = \frac{\pi}{4}$

$f(x) = \sin^{-1}(-1) + \tan^{-1}(-1) + \sec^{-1}(-1)$

$= \frac{\pi}{2} + \frac{\pi}{4} + 0$

$= \frac{3\pi}{4}$

13. The range of the function, $f(x) = \cot^{-1}\log_{0.5}\left(x^4 - 2x^2 + 3\right)$ is :

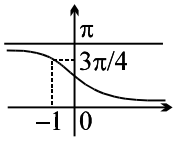
फलन $f(x) = \cot^{-1}\log_{0.5}\left(x^4 - 2x^2 + 3\right)$ का परिसर होगा :

(A) $(0, \pi)$ (B) $\left(0, \frac{3\pi}{4}\right]$ (C) $\left[\frac{3\pi}{4}, \pi\right)$ (D) $\left[\frac{\pi}{2}, \frac{3\pi}{4}\right]$

Ans. C

Sol. $y = (x^2 - 1)^2 + 2 \Rightarrow \quad y_{min} = 2$

$\Rightarrow \log_{0.5}(x^4 - 2x^2 + 3) \leq -1 \Rightarrow$ range $\left[\frac{3\pi}{4}, \pi\right)$



14. The range of the function $y = \left(\frac{\cos^{-1}(3x-1)}{\pi} + 1\right)^2$ is :

फलन $y = \left(\frac{\cos^{-1}(3x-1)}{\pi} + 1\right)^2$ का परिसर होगा :

(A) [1,4] (B) $[0,\pi]$ (C) $[1,\pi]$ (D) $[0,\pi^2]$

Ans. A

Sol. $\left(\frac{\cos^{-1}(3x-1)}{\pi} + 1\right)^2$

$\Rightarrow \left[\frac{[0,\pi]}{\pi} + 10\right]^2 = [[0,1] + 1]^2$

$= [1, 2]^2$

$= [1, 4]$

15. Let $f : R \to A = \left\{ y \mid 0 \le y < \frac{\pi}{2} \right\}$ be a function such that $f(x) = \tan^{-1}(x^2 + x + k)$, where k is a constant. The value of k for which f is an onto function, is :

(A) 1 (B) 0 (C) 1/4 (D) None of these

माना $f : R \to A = \left\{ y \mid 0 \le y < \frac{\pi}{2} \right\}$ एक फलन है जहाँ $f(x) = \tan^{-1}(x^2 + x + k)$, (k एक अचर है)। यदि f एक आच्छादक फलन है तो k होगा :

(A) 1 (B) 0 (C) 1/4 (D) इनमें से कोई नहीं

Ans. C

Sol. $f(x) = \tan^{-1}\left(x^2 + x + k\right)$

Given f(x) is onto

$\Rightarrow$ R(f) = co-domain $= \left[0, \frac{\pi}{2}\right)$

$\Rightarrow x^2 + x + k \ge 0 \ \forall \ x \in R$

$\Rightarrow D \le 0$

$\Rightarrow 1 - 4k \le 0$

$\Rightarrow k \ge \frac{1}{4}$

If $k = \frac{1}{4}$ then f is onto.

16. Domain of $f(x) = \sqrt{\sin^{-1}(2x) + \frac{\pi}{6}}$ is :

फलन $f(x) = \sqrt{\sin^{-1}(2x) + \frac{\pi}{6}}$ का प्रान्त है :

(A) $\left(-\frac{1}{2}, \frac{1}{2}\right]$ (B) $\left[-\frac{1}{4}, \frac{3}{4}\right)$ (C) $\left[-\frac{1}{4}, \frac{1}{4}\right]$ (D) $\left[-\frac{1}{4}, \frac{1}{2}\right]$

Ans. D

Sol. For $f(x)=\sqrt{\sin^{-1}(2x)+\frac{\pi}{6}}$ to be defined and real $\sin^{-1}2x + \pi/6 \geq 0$

$\Rightarrow \quad \sin^{-1} 2x \geq -\frac{\pi}{6}$ _____(1)

But we know that $-\pi/2 \leq \sin^{-1} 2x \leq \pi/2$ _____(2)

Combining (1) and (2), $-\pi/6 \leq \sin^{-1} 2x \leq \pi/2$

$\Rightarrow \sin(-\pi/6) \leq 2x \leq \sin(\pi/2) \Rightarrow -1/2 \leq 2x \leq 1$

$\Rightarrow \quad -1/4 \leq x \leq 1/2$

$\therefore \quad D_f = \left[-\frac{1}{4},\frac{1}{2}\right]$

17. Number of real value of x satisfying the equation, $\arctan\sqrt{x(x+1)} + \arcsin\sqrt{x(x+1)+1} = \frac{\pi}{2}$ is :

(A) 0 (B) 1 (C) 2 (D) more than 2

समीकरण $\tan\sqrt{x(x+1)} + \arcsin\sqrt{x(x+1)+1} = \frac{\pi}{2}$ को संतुष्ट करने वाले वास्तविक x की कुल संख्या कितनी होगी :

(A) 0 (B) 1 (C) 2 (D) 2 से अधिक

Ans. C

Sol. $x(x + 1) \geq 0$ and $0 \leq x^2 + x + 1 \leq 1$

$\Rightarrow x \geq 0$ or $x \leq -1$ and $x(x + 1) \leq 0$

$x \leq 0$ or $x \geq -1$

Hence $x = 0$ or $x = -1$

1. The value of $\tan\left(\cos^{-1}\frac{4}{5}+\tan^{-1}\frac{2}{3}\right)$ is :

(A) $\frac{17}{5}$ (B) $\frac{17}{6}$ (C) $\frac{1}{17}$ (D) None of these

$\tan\left(\cos^{-1}\frac{4}{5}+\tan^{-1}\frac{2}{3}\right)$ का मान होगा :

(A) $\frac{17}{5}$ (B) $\frac{17}{6}$ (C) $\frac{1}{17}$ (D) इनमें से कोई नहीं

Ans. B

Sol. $\tan\left(\cos^{-1}\frac{4}{5}+\tan^{-1}\frac{2}{3}\right)$

$\tan\left[\tan^{-1}\left(\frac{3}{4}\right)+\tan^{-1}\left(\frac{2}{3}\right)\right]$

$\tan\left[\tan^{-1}\left(\frac{\frac{3}{4}+\frac{2}{3}}{1-\frac{1}{2}}\right)\right]$

$=\frac{17}{6}$

2. The value of $\cos\left(2\cos^{-1}x+\sin^{-1}x\right)$ at $x=\frac{1}{5}$ is :

(A) $-\frac{2\sqrt{6}}{5}$ (B) $-2\sqrt{6}$ (C) $-\sqrt{\frac{6}{5}}$ (D) None of these

$\cos\left(2\cos^{-1}x+\sin^{-1}x\right)$ at $x=\frac{1}{5}$ का मान होगा :

(A) $-\frac{2\sqrt{6}}{5}$ (B) $-2\sqrt{6}$ (C) $-\sqrt{\frac{6}{5}}$ (D) इनमें से कोई नहीं

Ans. A

Sol. $\cos\left(2\cos^{-1}x + \sin^{-1}x\right)$

$$\cos\left[2\cos^{-1}x + \frac{\pi}{2} - \cos^{-1}x\right]$$

$$\Rightarrow \cos\left[\frac{\pi}{2} + \cos^{-1}x\right]$$

$$\Rightarrow -\sin\left[\cos^{-1}x\right] = -\sin\left(\cos^{-1}\left(\frac{1}{5}\right)\right)$$

$$-\frac{2\sqrt{6}}{5}$$

3. The value of $\sin\left(2\tan^{-1}\frac{1}{3}\right) + \cos\left(\tan^{-1}2\sqrt{2}\right)$ is :

(A) $\frac{14}{15}$ (B) $\frac{15}{14}$ (C) $\frac{1}{15}$ (D) None of these

$\sin\left(2\tan^{-1}\frac{1}{3}\right) + \cos\left(\tan^{-1}2\sqrt{2}\right)$ का मान होगा :

(A) $\frac{14}{15}$ (B) $\frac{15}{14}$ (C) $\frac{1}{15}$ (D) इनमें से कोई नहीं

Ans. A

Sol. $\sin\left(2\tan^{-1}\frac{1}{3}\right) + \cos\left(\tan^{-1}2\sqrt{2}\right)$

$$\tan^{-1}\frac{1}{3} = \theta$$

$$\tan\theta = \frac{1}{3} \Rightarrow \sin 2\theta = \frac{2\tan\theta}{1+\tan^{-1}\theta} = \frac{\frac{2}{3}}{1+\frac{1}{4}} = \frac{\frac{2}{3}}{\frac{5}{3}} = \frac{3}{5}$$

$$\tan^{-1}2\sqrt{2} = \phi$$

$$\tan\phi = \frac{2\sqrt{2}}{1}$$

$$\cos\phi = \frac{1}{3}$$

$$\sin 2\theta + \cos\phi = \frac{3}{5} + \frac{1}{3} = \frac{9+5}{15}$$

$$= \frac{14}{15}$$

4. The value of $\tan\left\{4\ \tan^{-1}\left(\frac{1}{5}\right) - \tan^{-1}\left(\frac{1}{239}\right)\right\}$ is :

$\tan\left\{4\ \tan^{-1}\left(\frac{1}{5}\right) - \tan^{-1}\left(\frac{1}{239}\right)\right\}$ का मान होगा :

(A) $\frac{1}{\sqrt{2}}$ (B) $\frac{1}{\sqrt{3}}$ (C) $\sqrt{3}$ (D) 1

Ans. D

Sol. $\tan\left\{4\ \tan^{-1}\left(\frac{1}{5}\right) - \tan^{-1}\left(\frac{1}{239}\right)\right\}$

$$= \tan\left[2\tan^{-1}\left(\frac{1}{5}\right) - 2\tan^{-1}\left(\frac{1}{5}\right) - \tan^{-1}\left(\frac{1}{239}\right)\right]$$

$$= \tan\left[\tan^{-1}\left(\frac{\frac{2}{5}}{1-\frac{1}{25}}\right) + \tan^{-1}\left(\frac{\frac{2}{5}}{1-\frac{1}{\sqrt{3}}}\right) - \tan^{-1}\left(\frac{1}{239}\right)\right]$$

$$= \tan\left[\tan^{-1}\left(\frac{2+5}{12}\right) + \tan^{-1}\left(\frac{5}{12}\right) - \tan^{-1}\left(\frac{1}{239}\right)\right]$$

$$= \tan\left[\tan^{-1}\left(\frac{\frac{2\times 5}{12}}{1-\frac{25}{144}}\right) - \tan^{-1}\left(\frac{1}{239}\right)\right]$$

$$\Rightarrow \tan\left[\tan^{-1}\left(\frac{10\times12}{119}\right)-\tan^{-1}\left(\frac{1}{239}\right)\right]$$

$$\Rightarrow \tan\left[\tan^{-1}\left(\frac{\frac{120}{119}-\frac{1}{239}}{1+\frac{120}{119}\times\frac{1}{239}}\right)\right]$$

$$=\tan\left[\tan^{-1}\left(\frac{120\times239-119}{119\times239+120}\right)\right]$$

$$\Rightarrow \tan\left[\tan^{-1}\left(\frac{28,561}{28561}\right)\right]$$

$= \tan(\tan^{-1}(1))$

$= 1$

5. If $A=\sin^{-1}\left\{\cot\left(\sin^{-1}\sqrt{\frac{13}{17}}\right)\right\}$ then :

(A) $A=\tan^{-1}\frac{2}{3}$ (B) $A>\tan^{-1}\frac{2}{3}$ (C) $A<\tan^{-1}\frac{2}{3}$ (D) None of these

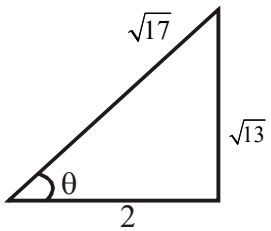
यदि $A=\sin^{-1}\left\{\cot\left(\sin^{-1}\sqrt{\frac{13}{17}}\right)\right\}$ तब :

(A) $A=\tan^{-1}\frac{2}{3}$ (B) $A>\tan^{-1}\frac{2}{3}$ (C) $A<\tan^{-1}\frac{2}{3}$ (D) इनमें से कोई नहीं

Ans. A

Sol. $A=\sin^{-1}\left\{\cot\left(\sin^{-1}\sqrt{\frac{13}{17}}\right)\right\}$

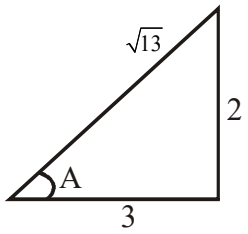
$\sin\theta=\frac{\sqrt{13}}{\sqrt{17}}$



$$\cot\theta = \frac{2}{\sqrt{3}}$$

$$A = \sin^{-1}\left(\frac{2}{\sqrt{13}}\right)$$

$$\sin A = \frac{2}{\sqrt{13}}$$



$$\tan A = \frac{2}{3}$$

6. The value of $\cos\left(\frac{1}{2}\cos^{-1}\frac{1}{8}\right)$ is equal to :

$\cos\left(\frac{1}{2}\cos^{-1}\frac{1}{8}\right)$ का मान बराबर होगा :

(A) 3/4 (B) –3/4 (C) 1/16 (D) 1/4

Ans. A

Sol. $\cos\left(\frac{1}{2}\cos^{-1}\frac{1}{8}\right)$

$$\cos^{-1}\frac{1}{8} = \theta$$

$$\cos\theta = \frac{1}{8}$$

$$\Rightarrow \cos\left(\frac{\theta}{2}\right) = ?$$

$$\cos\frac{\theta}{2} \le \sqrt{\frac{1+\cos\theta}{2}}$$

$$= \sqrt{\frac{1+\frac{1}{8}}{2}} = \frac{3}{4}$$

7. If $\cot^{-1}\frac{n}{\pi} > \frac{\pi}{6}$, $n \in N$, then the maximum value of n is :

(A) 1 (B) 5 (C) 9 (D) none of these

यदि $\cot^{-1}\frac{n}{\pi} > \frac{\pi}{6}$, $n \in N$, तब n का अधिकतम मान होगा :

(A) 1 (B) 5 (C) 9 (D) इनमें से कोई नहीं

Ans. B

Sol. $\cot^{-1}\frac{n}{\pi} > \frac{\pi}{6} \quad \Rightarrow \quad -\infty < \frac{n}{\pi} < \cot\frac{\pi}{6}$

$\Rightarrow n < \pi\sqrt{3} \quad n \in N \ , \quad n_{max} = 5$

8. If $17x^2 + 17x\left(\tan\left(2\tan^{-1}\left(\frac{1}{5}\right) - \frac{\pi}{4}\right)\right) - 10 = 0$, then 'x' is equal to :

यदि $17x^2 + 17x\left(\tan\left(2\tan^{-1}\left(\frac{1}{5}\right) - \frac{\pi}{4}\right)\right) - 10 = 0$, तो x बराबर होगा :

(A) $\frac{10}{17}, 1$ (B) $-1, \frac{10}{17}$ (C) $\frac{7}{17}, -1$ (D) $1 , -\frac{10}{17}$

Ans. D

Sol. $17x^2 + 17x\left(\tan\left(2\tan^{-1}\left(\frac{1}{5}\right) - \frac{\pi}{4}\right)\right) - 10 = 0$

$$17x^2 + 17x\left[\tan\left(\tan^{-1}\frac{\frac{2}{5}}{1-\frac{1}{25}} - \frac{\pi}{4}\right)\right] - 10 = 0$$

$$17x^2 + 17x\left[\tan\left[\tan^{-1}\left(\frac{5}{12}\right) - \tan^{-1}(1)\right]\right] - 10 = 0$$

$$17x^2+17x\left[\tan\left[\tan^{-1}\left(\frac{\frac{5}{12}-1}{1+\frac{5}{12}}\right)\right]\right]-10=0$$

$$17x^2+17x\left(\frac{-\frac{7}{12}}{\frac{14}{14}}\right)-10=0$$

$17x^2 - 7x - 10 = 0$

$17x^2 - 17x + 10x - 10 = 0$

$17x(x-1) + 10(x-1) = 0$

$$x = -\frac{10}{17}, +1$$

9. The value of the angle $\tan^{-1}(\tan 65° - 2\tan 40°)$ in degrees is equal to :

$\tan^{-1}(\tan 65° - 2\tan 40°)$ का मान डिग्री (degrees) में होगा :

(A) $-20°$ (B) $20°$ (C) $25°$ (D) $40°$

Ans. C

Sol. Consider

$\tan 65° - 2\tan 40°$

$\tan(45° + 20°) - 2\tan 40°$

$$\frac{1+\tan 20°}{1-\tan 20°} - \frac{4\tan 20°}{1-\tan^2 20°}$$

$$\frac{(1+\tan 20°)^2 - 4\tan 20°}{(1-\tan 20°)(1+\tan 20°)} = \frac{(1-\tan 20°)(1-\tan 20°)}{(1-\tan 20°)(1+\tan 20°)} = \frac{(1-\tan 20°)}{(1+\tan 20°)}$$

$= \tan(45° - 20°) = \tan 25°$

$\therefore$ $\tan^{-1}(\tan 25°) = 25°$ Ans.

10. $\tan\left(\frac{\pi}{4} + \frac{1}{2}\cos^{-1} x\right) + \tan\left(\frac{\pi}{4} - \frac{1}{2}\cos^{-1} x\right)$, $x \neq 0$, is equal to :

(A) x (B) 2x (C) $\frac{2}{x}$ (D) none of these

$\tan\left(\frac{\pi}{4}+\frac{1}{2}\cos^{-1}x\right)+\tan\left(\frac{\pi}{4}-\frac{1}{2}\cos^{-1}x\right)$, $x \neq 0$, बराबर होगा :

(A) x (B) 2x (C) $\frac{2}{x}$ (D) इनमें से कोई नहीं

Ans. C

Sol. $\tan\left(\frac{\pi}{4}+\frac{1}{2}\cos^{-1}x\right)+\tan\left(\frac{\pi}{4}-\frac{1}{2}\cos^{-1}x\right)\ x \neq 0$

let (माना) $\theta=\frac{1}{2}\cos^{-1}x \qquad 2\theta \in [0,\pi]-\left\{\frac{\pi}{2}\right\}$

$$=\tan\left(\frac{\pi}{4}+\theta\right)+\tan\left(\frac{\pi}{4}-\theta\right)=\frac{1+\tan\theta}{1-\tan\theta}+\frac{1-\tan\theta}{1+\tan\theta}=2\left(\frac{1+\tan^2\theta}{1-\tan^2\theta}\right)=\frac{2}{\cos 2\theta}=\frac{2}{\cos\cos^{-1}x}=\frac{2}{x}$$

11. If $-1 < x < 0$ then $\sin^{-1} x$ equals :

यदि $-1 < x < 0$ तो $\sin^{-1} x$ बराबर होगा :

(A) $\pi - \cos^{-1}\left(\sqrt{1-x^2}\right)$ (B) $\tan^{-1}\left(\frac{x}{\sqrt{1-x^2}}\right)$

(C) $-\cot^{-1}\left(\frac{\sqrt{1-x^2}}{x}\right)$ (D) $\text{cosec}^{-1} x$

Ans. B

Sol. $-1 < x < 0$

$\sin^{-1}(x)$

$\tan^{-1}\left(\frac{x}{\sqrt{1-x^2}}\right)$

12. The equation $3\sin^{-1} x = \sin^{-1}(3x - 4x^3)$ is satisfied by :

(A) $-\frac{1}{2} \le x \le \frac{1}{2}$ (B) $0 \le x \le 1$ (C) $x \ge \frac{1}{2}$ (D) None of these

समीकरण $3\sin^{-1} x = \sin^{-1}(3x - 4x^3)$ निम्न में से किसके लिए सन्तुष्ट होती है :

(A) $-\frac{1}{2} \le x \le \frac{1}{2}$ (B) $0 \le x \le 1$ (C) $x \ge \frac{1}{2}$ (D) इनमें से कोई नहीं

Ans. A

Sol. $3\sin^{-1}x = \sin^{-1}(3x - 4x^3)$

$x \in [-1,1]$ $\quad -1 \le 3x - 4x^3 \le 1$

$3\sin^{-1}x = \left[\frac{-\pi}{2}, \frac{\pi}{2}\right]$

$\sin^{-1}x = \left[\frac{-\pi}{6}, \frac{\pi}{6}\right] \Rightarrow x \in \left[\frac{-1}{2}, \frac{1}{2}\right]$

13. The smallest and the largest values of $\tan^{-1}\left(\frac{1-x}{1+x}\right), 0 \le x \le 1$ are :

$\tan^{-1}\left(\frac{1-x}{1+x}\right), 0 \le x \le 1$, का न्यूनतम व अधिकतम मान होगा :

(A) $0, \pi$ (B) $0, \frac{\pi}{4}$ (C) $-\frac{\pi}{4}, \frac{\pi}{4}$ (D) $\frac{\pi}{4}, \frac{\pi}{2}$

Ans. B

Sol. $\tan^{-1}\left(\frac{1-x}{1+x}\right), 0 \le x \le 1$

$x = 1 \;\; x = 1 \Rightarrow \tan^{-1}(0) = 0$

$x = 0 \; \tan^{-1}(1) = \frac{\pi}{4}$

1. The value of $\tan\{\sin^{-1}(\cos(\sin^{-1}x))\} \cdot \tan\{\cos^{-1}(\sin(\cos^{-1}x))\}; x \in (0, 1)$ is equal to :
(A) 0 (B) 1 (C) –1 (D) None of these

यदि $x \in (0, 1)$ हो, तो $\tan\{\sin^{-1}(\cos(\sin^{-1}x))\} \cdot \tan\{\cos^{-1}(\sin(\cos^{-1}x))\}$; का मान बराबर होगा :
(A) 0 (B) 1 (C) –1 (D) इनमें से कोई नहीं

Ans. B

Sol. $\tan\left[\sin^{-1}\left(\cos(\sin^{-1}x)\right)\right] \quad \tan\left[\cos^{-1}\left(\sin(\cos^{-1}x)\right)\right]$

$\sin^{-1}x = \theta \qquad \cos^{-1}x = \phi$

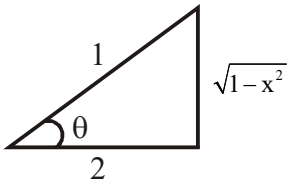
$\sin\theta = \frac{x}{1} \qquad \cos\theta = x$

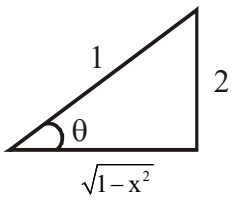
$\cos\theta = \sqrt{1-x^2} \qquad \sin\theta = \sqrt{1-x^2}$

$\Rightarrow \tan\left[\sin^{-1}\left(\sqrt{1-x^2}\right)\right] \tan\left[\cos^{-1}\left(\sqrt{1-x^2}\right)\right]$

$\sin^{-1}\sqrt{1-x^2} = A \qquad \cos^{-1}\sqrt{1-x^2} = B$

$\sin A = \frac{\sqrt{1-x^2}}{1} \qquad \cos B = \sqrt{1-x^2}$





$\tan A = \frac{\sqrt{1-x^2}}{2} \qquad \tan B = \frac{x}{\sqrt{1-x^2}}$

$\tan A \cdot \tan B = 1$

2. The range of the function, $f(x) = \cot^{-1}x + \sec^{-1}x + \text{cosec}^{-1}x$, is :

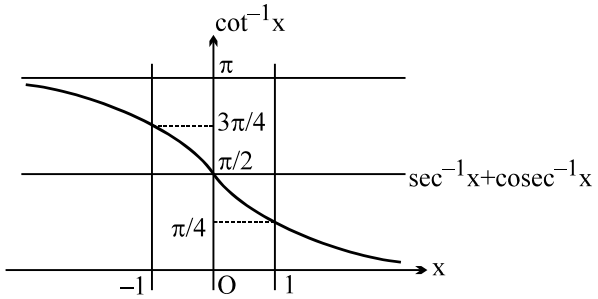
फलन $f(x) = \cot^{-1}x + \sec^{-1}x + \text{cosec}^{-1}x$ का परिसर होगा :

(A) $\left(\frac{\pi}{2}, \frac{3\pi}{2}\right)$ (B) $\left(\frac{\pi}{2}, \frac{3\pi}{4}\right] \cup \left[\frac{5\pi}{4}, \frac{3\pi}{2}\right)$

(C) $\left[\frac{\pi}{2}, \pi\right) \cup \left(\pi, \frac{3\pi}{2}\right]$ (D) $\left(\frac{\pi}{2}, \pi\right) \cup \left(\pi, \frac{3\pi}{2}\right)$

Ans. B

Sol.



Domain of $f(x)$ is $|x| \geq 1$

From the graph for $x \geq 1$ $f(x)$ can attain values from $\left(\frac{\pi}{2}, \frac{3\pi}{4}\right]$

also for $x \leq -1$, $f(x)$ can attain values from $\left[\frac{5\pi}{4}, \frac{3\pi}{2}\right)$

3. If $a \leq \sin^{-1}x + \cos^{-1}x + \tan^{-1}x \leq b$, then :

(A) $a = \frac{\pi}{4}, b = \frac{3\pi}{4}$ (B) $a = 0,\ b = \pi$ (C) $a = -\frac{\pi}{4}, b = \frac{3\pi}{4}$ (D) None of these

यदि $a \leq \sin^{-1}x + \cos^{-1}x + \tan^{-1}x \leq b$, तब :

(A) $a = \frac{\pi}{4}, b = \frac{3\pi}{4}$ (B) $a = 0,\ b = \pi$ (C) $a = -\frac{\pi}{4}, b = \frac{3\pi}{4}$ (D) इनमें से कोई नहीं

Ans. A

Sol. $a \leq \sin^{-1}x + \cos^{-1}x + \tan^{-1}x \leq b$

$$a \leq \frac{\pi}{2} + \tan^{-1}x \leq b$$

$$a \leq \frac{\pi}{2} + \left[\frac{-\pi}{4}, \frac{\pi}{4}\right] \leq b$$

$\left[\frac{\pi}{4}, \frac{3\pi}{4}\right]$

$a = \frac{\pi}{4}$ $\quad\quad$ $b = \frac{3\pi}{4}$

4. Complete solution set of inequation $\left(\cos^{-1} x\right)^2 - \left(\sin^{-1} x\right)^2 > 0$, is :

(A) $\left[0 , \frac{1}{\sqrt{2}}\right)$ (B) $\left[-1 , \frac{1}{\sqrt{2}}\right)$ (C) $\left(-1 , \sqrt{2}\right)$ (D) None of these

असमिका $\left(\cos^{-1} x\right)^2 - \left(\sin^{-1} x\right)^2 > 0$ का सम्पूर्ण हल समुच्चय होगा :

(A) $\left[0 , \frac{1}{\sqrt{2}}\right)$ (B) $\left[-1 , \frac{1}{\sqrt{2}}\right)$ (C) $\left(-1 , \sqrt{2}\right)$ (D) इनमें से कोई नहीं

Ans. B

Sol. $\left(\cos^{-1} x\right)^2 - \left(\sin^{-1} x\right)^2 > 0$

$\left(\cos^{-1} x + \sin^{-1} x\right)\left(\cos^{-1} x - \sin^{-1} x\right) > 0$

$\left(\frac{\pi}{2}\right)\left(\cos^{-1} x - \sin^{-1} x\right) > 0$

$\frac{\cos^{-1} x}{[0,\pi]} > \frac{\sin^{-1}}{\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]}$

$x \in \left[-1 , \frac{1}{\sqrt{2}}\right)$

5. If $\sin^{-1}\left(x - \frac{x^2}{2} + \frac{x^3}{4} - .....\right) + \cos^{-1}\left(x^2 - \frac{x^4}{2} + \frac{x^6}{4} - .....\right) = \frac{\pi}{2}$ for $0 < |x| < \sqrt{2}$ then x equals to :

यदि $\sin^{-1}\left(x - \frac{x^2}{2} + \frac{x^3}{4} - .....\right) + \cos^{-1}\left(x^2 - \frac{x^4}{2} + \frac{x^6}{4} - .....\right) = \frac{\pi}{2}$, $0 < |x| < \sqrt{2}$ तो x बराबर है :

(A) 1/2 (B) 1 (C) –1/2 (D) –1

Ans. B

Sol. $\sin^{-1}\left(x-\frac{x^2}{2}+\frac{x^3}{4}.....\right)+\cos^{-1}\left(x^2-\frac{x^4}{2}+\frac{x^6}{4}+...\right)=\frac{\pi}{2} \Rightarrow \sin^{-1}\left(\frac{x}{1+(x/2)}\right)+\cos^{-1}\left(\frac{x^2}{1+(x^2/2)}\right)=\frac{\pi}{2}$

$\frac{2x}{2+x}=\frac{2x^2}{2+x^2} \quad \Rightarrow \quad 2x+x^3=2x^2+x^3$

$x = 0, 1$ But किन्तु $\therefore\ |x| > 0$

so $x = 1$ is the only answer. अतः $x = 1$ एक मात्र हल है।

6. The number of solutions of the equation, $\cos^{-1}x - \sin^{-1}x = \cos^{-1}x\sqrt{3}$ is :

(A) 1 (B) 2 (C) 3 (D) None of these

समीकरण $\cos^{-1}x - \sin^{-1}x = \cos^{-1}x\sqrt{3}$ के कुल हलों की संख्या होगी :

(A) 1 (B) 2 (C) 3 (D) इनमें से कोई नहीं

Ans. C

Sol. $\cos^{-1}x - \sin^{-1}x = \cos^{-1}x\sqrt{3}$

$\cos^{-1}x = A \qquad \sin^{-1}x = B$

$\cos A = x \qquad \sin B = x$

$\sin A = \sqrt{1-x^2} \qquad \cos B = \sqrt{1-x^2}$

$A - B = \cos^{-1}\left(x\sqrt{3}\right)$

$\cos(A-B) = x\sqrt{3}$

$\cos A\cos B + \sin A\sin B = x\sqrt{3}$

$x\sqrt{1-x^2}+\sqrt{1-x^2}(x) = x\sqrt{3}$

$2x\sqrt{1-x^2} = x\sqrt{3}$

$x = 0 \quad 4(1-x^2) = 3$

$x = \frac{1}{4} \Rightarrow x = \pm\frac{1}{2}$

7. If $\sin^{-1}\frac{x}{5}+\sec^{-1}\left(\frac{5}{4}\right)=\frac{\pi}{2}$, then x is equal to :

यदि $\sin^{-1}\frac{x}{5}+\sec^{-1}\left(\frac{5}{4}\right)=\frac{\pi}{2}$ तब x बराबर होगा :

(A) 4 (B) 5 (C) 1 (D) 3

Ans. A

Sol. $\sin^{-1}\frac{x}{5}+\sec^{-1}\left(\frac{5}{4}\right)=\frac{\pi}{2}$

$\sin^{-1}\left(\frac{x}{5}\right)+\cos^{-1}\left(\frac{4}{5}\right)=\frac{\pi}{2}$

$x=4$

8. If $\sin^{-1}x+\sin^{-1}2x=\frac{\pi}{3}$, then x equals :

(A) $\frac{1}{2}\sqrt{\frac{3}{7}}$ (B) $\frac{1}{2}\sqrt{\frac{3}{13}}$ (C) $\frac{1}{2}\sqrt{\frac{7}{3}}$ (D) None of these

यदि $\sin^{-1}x+\sin^{-1}2x=\frac{\pi}{3}$ तब x बराबर होगा :

(A) $\frac{1}{2}\sqrt{\frac{3}{7}}$ (B) $\frac{1}{2}\sqrt{\frac{3}{13}}$ (C) $\frac{1}{2}\sqrt{\frac{7}{3}}$ (D) इनमें से कोई नहीं

Ans. A

Sol. $\sin^{-1}x+\sin^{-1}2x=\frac{\pi}{3}$

$\sin^{-1}x=A$ $\sin^{-1}2x=B$

$\sin A=x$ $\sin B=2x$

$\cos A=\sqrt{1-x^2}$ $\cos B=\sqrt{1-4x^2}$

$A+B=\frac{\pi}{3}$

$\cos(A+B)=\cos\frac{\pi}{3}$

$\cos A\cos B-\sin A\sin B=\frac{1}{2}$

$\sqrt{1-x^2}\sqrt{1-4x^2}-2x^2=\frac{1}{2}$

$$\sqrt{(1-x^2)(1-4x^2)} = \frac{1+4x^2}{2}$$

$$1 - 5x^2 + 4x^2 = \frac{1+16x^2+8x^2}{4}$$

$$4 - 20x^2 + 16x^4 = 1 + 16x^4 + 8x^2$$

$$28x^2 = 3$$

$$x^2 = \frac{3}{29}$$

$$x = \frac{1}{2}\sqrt{\frac{3}{7}}$$

9. The number of solutions of the equation $\tan^{-1}\left(\frac{1}{2x+1}\right) + \tan^{-1}\left(\frac{1}{4x+1}\right) = \tan^{-1}\left(\frac{2}{x^2}\right)$ is :

समीकरण $\tan^{-1}\left(\frac{1}{2x+1}\right) + \tan^{-1}\left(\frac{1}{4x+1}\right) = \tan^{-1}\left(\frac{2}{x^2}\right)$ के हलों की संख्या होगी :

(A) 0 (B) 1 (C) 2 (D) 3

Ans. B

Sol. $x = 3$ is the only solution, $x = 0$ and $x = -2/3$ is rejected

10. Number of value of x satisfying the equation $\sin^{-1}\left(\frac{5}{x}\right) + \sin^{-1}\left(\frac{12}{x}\right) = \frac{\pi}{2}$ is :

(A) 0 (B) 1 (C) 2 (D) more than 2

समीकरण $\sin^{-1}\left(\frac{5}{x}\right) + \sin^{-1}\left(\frac{12}{x}\right) = \frac{\pi}{2}$ को संतुष्ट करने वाले कुल x की संख्या होगी :

(A) 0 (B) 1 (C) 2 (D) 2 से अधिक

Ans. B

Sol. $\sin^{-1}\left(\frac{5}{x}\right) = \frac{\pi}{2} - \sin^{-1}\left(\frac{12}{x}\right) = \cos^{-1}\left(\frac{12}{x}\right)$ (very obviously $x > 0$)

$\alpha = \beta$ where $\sin^{-1}\frac{5}{x} = \alpha$ and $\cos^{-1}\frac{12}{x} = \beta$

$\tan\alpha = \tan\beta$

$$\frac{5}{\sqrt{x^2-25}} = \frac{\sqrt{x^2-144}}{12}$$

solving $x = 13$ or $x = -13$ (rejected)

11. The Sum $\sum_{n=1}^{\infty}\tan^{-1}\left(\frac{2}{n^2+n+4}\right)$ is equal to :

$\sum_{n=1}^{\infty}\tan^{-1}\left(\frac{2}{n^2+n+4}\right)$ का मान होगा :

(A) $\tan^{-1}2$ (B) $\frac{\pi}{2} + \tan^{-1}2$ (C) $\frac{\pi}{2} - \tan^{-1}2$ (D) $\frac{\pi}{4}$

Ans. A

Sol. $$\sum_{n=1}^{\infty}\tan^{-1}\left(\frac{2}{n^2+n+4}\right) = \sum_{n=1}^{\infty}\tan^{-1}\left\{\frac{\frac{2}{n(n+1)}}{1+\frac{4}{n(n+1)}}\right\}$$

$$= \sum_{n=1}^{\infty}\tan^{-1}\left\{\frac{\frac{2}{n}-\frac{2}{n+1}}{1+\frac{2}{n}\cdot\frac{2}{n+1}}\right\} = \sum_{n=1}^{\infty}\left\{\tan^{-1}\left(\frac{2}{n}\right)-\tan^{-1}\left(\frac{2}{n+1}\right)\right\}$$

$$= \left\{\tan^{-1}\left(\frac{2}{1}\right)-\tan^{-1}\left(\frac{2}{2}\right)\right\}+\left\{\tan^{-1}\left(\frac{2}{2}\right)-\tan^{-1}\left(\frac{2}{3}\right)\right\}+.....\infty \text{ terms}$$

$$= \tan^{-1}2$$

12. The value of $\sum_{n=1}^{\infty}\left(\tan^{-1}\left(\frac{n}{n+2}\right)-\tan^{-1}\left(\frac{n-1}{n+1}\right)\right)$ is equal to :

$\sum_{n=1}^{\infty}\left(\tan^{-1}\left(\frac{n}{n+2}\right)-\tan^{-1}\left(\frac{n-1}{n+1}\right)\right)$ का मान होगा :

(A) $\frac{\pi}{4}$ (B) $\frac{\pi}{3}$ (C) $\frac{\pi}{2}$ (D) $\frac{3\pi}{4}$

Ans. A

Sol. We have $T_1 = \tan^{-1}\frac{1}{3} - \tan^{-1}0$

$$T_2 = \tan^{-1}\frac{1}{2} - \tan^{-1}\frac{1}{3}$$

$$T_3 = \tan^{-1}\frac{3}{5} - \tan^{-1}\frac{1}{2}$$

$$\vdots \qquad \vdots \qquad \vdots$$

$$T_n = \tan^{-1}\left(\frac{n}{n+2}\right) - \tan^{-1}\left(\frac{n-1}{n+1}\right)$$

$\therefore$ On adding all above equation, we get $S_n = \sum_{r=1}^{n} T_r = \tan^{-1}\left(\frac{n}{n+2}\right)$

Hence $S_\infty = \underset{n\to\infty}{\text{Limit}} \tan^{-1}\left(\frac{n}{n+2}\right) = \tan^{-1}1 = \frac{\pi}{4}$ Ans.

निम्नलिखित श्रेणीयों का योग ज्ञात कीजिए :

13. $\cot^{-1}\left(2^2 + \frac{1}{2}\right) + \cot^{-1}\left(2^3 + \frac{1}{2^2}\right) + \cot^{-1}\left(2^4 + \frac{1}{2^3}\right) + ... \infty.$

Ans. $\tan^{-1} 1/2$

Sol. $\tan^{-1}\frac{1}{3} + \tan^{-1}\frac{2}{9} + ... + \tan^{-1}\frac{2^{n-1}}{1+2^{2n-1}} + ... + \text{upto infinite terms (अनन्त पदों तक)}$

$$= \sum_{n=1}^{\infty}\tan^{-1}\frac{2^{n-1}}{1+2^{2n-1}} = \sum_{n=1}^{\infty}\tan^{-1}\frac{2^n - 2^{n-1}}{1+2^n.2^{n-1}} = \sum_{n=1}^{\infty}\{\tan^{-1}2^n - \tan^{-1}2^{n-1}\}$$

$$= [(\tan^{-1}2 - \tan^{-1}1) + (\tan^{-1}2^2 - \tan^{-1}2) + (\tan^{-1}2^3 - \tan^{-1}2^2) + ... + \underset{n\to\infty}{\ell\text{im}}\,(\tan^{-1}2^n - \tan^{-1}2^{n-1})]$$

$$= \underset{n\to\infty}{\ell\text{im}}\,\tan^{-1}2^n - \tan^{-1}1 = \frac{\pi}{2} - \frac{\pi}{4} = \frac{\pi}{4}$$

14. $\text{cosec}^{-1}\sqrt{5} + \text{cosec}^{-1}\sqrt{65} + \text{cosec}^{-1}\sqrt{325} + \text{cosec}^{-1}\sqrt{1025} + \text{cosec}^{-1}\sqrt{2501} + ... + \infty$

Ans. $\pi/4$

Sol. $\text{cosec}^{-1}\sqrt{5} + \text{cosec}^{-1}\sqrt{65} + \text{cosec}^{-1}\sqrt{325} + \text{cosec}^{-1}\sqrt{1025} + \text{cosec}^{-1}\sqrt{2501} + ... + \infty$

$$\Rightarrow \tan^{-1}\frac{1}{2} + \tan^{-1}\frac{1}{8} + \tan^{-1}\left(\frac{1}{18}\right) + \tan^{-1}\left(\frac{1}{32}\right) + \tan^{-1}\left(\frac{1}{50}\right) + ...$$

$$T_r = \tan^{-1}\left(\frac{1}{2r^2}\right) = \tan^{-1}\left(\frac{2}{1+4r^2-1}\right)$$

$$T_r = \tan^{-1}\left[\frac{2}{1+(2r-1)(2r+1)}\right]$$

$$= \tan^{-1}\left[\frac{(2r+1)-(2r-1)}{1+(2r-1)(2r+1)}\right]$$

$T_r = \tan^{-1}(2r+1) - \tan^{-1}(2r-1)$

$T_1 = \tan^{-1}(3) - \tan^{-1}(1)$

$T_2 = \tan^{-1}(5) - \tan^{-1}(3)$

$T_3 = \tan^{-1}(7) - \tan^{-1}(5)$

$T_4 = \tan^{-1}(9) - \tan^{-1}(7)$

$T_n = \tan^{-1}(2n+1) - \tan^{-1}(2n-1)$

$S_n = \tan^{-1}(2n+1) - \tan^{-1}(1)$

$S_\infty = \tan^{-1}(\infty) - \tan^{-1}(1)$

$$= \frac{\pi}{2} - \frac{\pi}{4}$$

$$S_\infty = \frac{\pi}{4}$$